

HAT-P-36b

R-1382

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-36_210418 · created 2026-08-01T02:03:10+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459322.9066 +/- 0.0027 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.126 +/- 0.011
Transit depth (Rp/Rs)^2	1.58 +/- 0.29 [%]
Orbital Inclination (inc)	85.96 +/- 2.5
Airmass coefficient 1 (a1)	1.08 +/- 0.012
Airmass coefficient 2 (a2)	-0.0489 +/- 0.0039
Scatter in the residuals of the lightcurve fit is	1.13 %
Best Comparison Star	#3 - [100, 82]
Optimal Aperture	4.34
Optimal Annulus	13.75
Transit Duration (day)	0.0925 +/- 0.006

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.126 ± 0.011 vs 0.126 (deviation 0.0σ)
Duration fit vs published	0.0925 d vs 0.0925 d (deviation 0.0σ)
Mid-time O–C	8.5 min
Depth SNR	10.3
Residual scatter	1.13 %

Light curve

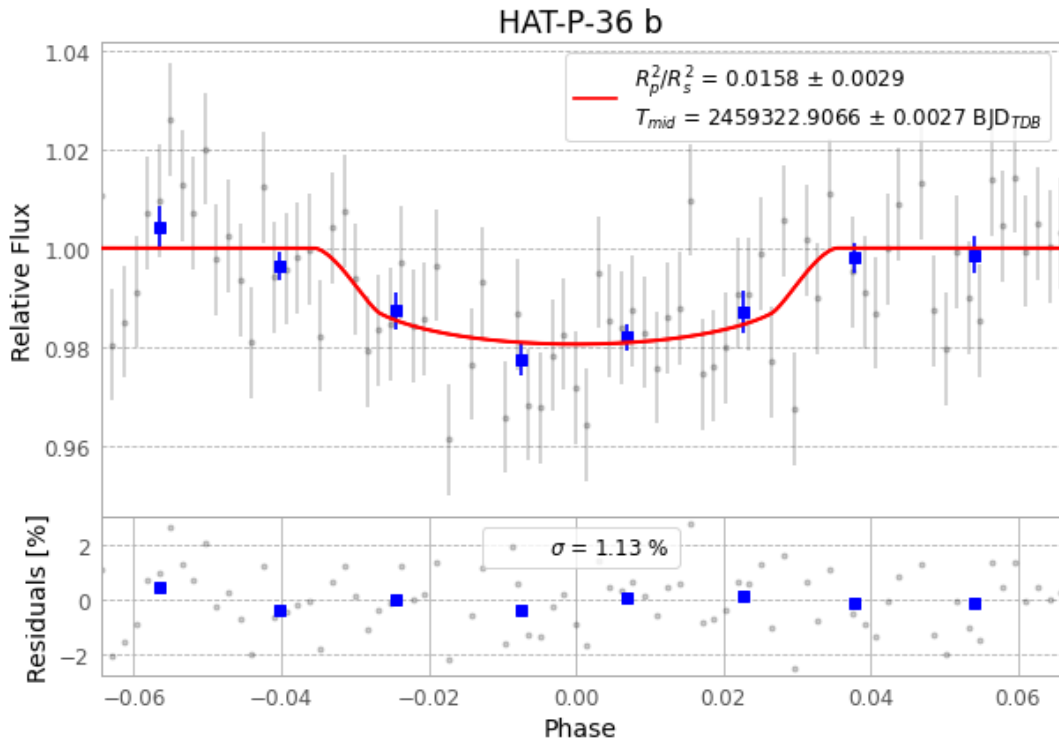


Figure 1 — FinalLightCurve_HAT-P-36 b_2021-04-18.png (SHA-256 4047d3cb295a61b1...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [405, 245], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 e05f7130d05ff03c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

84 FITS frames (55 MB), HATP-36210418073612.FITS → HATP-36210418114512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-210418.log (cc3bf290375a...), FinalLightCurve_HAT-P-36 b_2021-04-18.png (4047d3cb295a...), FinalLightCurve_HAT-P-36 b_2021-04-18.csv (8edbf70170a...)

Compute resources

Wall time 140.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 02:03Z.